

Coding & Solution

Date	01 July 2026
Team ID	ghousiyaunnissa@gmail.com
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/shaikghousunnissa123/Credit-card-approval-Prediction/tree/main
Programming Language(s)	Python 3.10
Framework(s) Used	Flask Microframework, Scikit-Learn Ecosystem
Key Features Implemented	• Interactive multi-field financial web application data form validation dashboard interface. • Continuous automated one-hot matrix categorical alignment utilizing structural templates via <code>model_columns.pkl</code>. • Real-time predictive analytics using an optimized Random Forest parallel voting ensemble classifier engine. • Instant user interface response updates containing structural green/red prediction outcome condition panels.
Pending / Incomplete Features	None. Core routing configurations, structural pipeline assets, and front-facing interface styles are fully checked in and functional.
Setup / Run Instructions	1. Install necessary dependencies: <code>pip install flask pandas scikit-learn</code> 2. Place <code>approval_model.pkl</code> and <code>model_columns.pkl</code> in the root execution environment directory. 3. Launch the native server app engine runtime framework by executing: <code>python app.py</code> 4. Open any standard local web browser window and route parameters directly to: <code>http://localhost:5000</code>

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The application source architecture securely isolates matrix transformation logic from server routing controls. Execution exceptions inside the prediction pipeline are gracefully caught via structural try-except handling routines to prevent runtime process crashing. All serialized ML dependencies maintain strict feature dimension synchronization across validation runs.